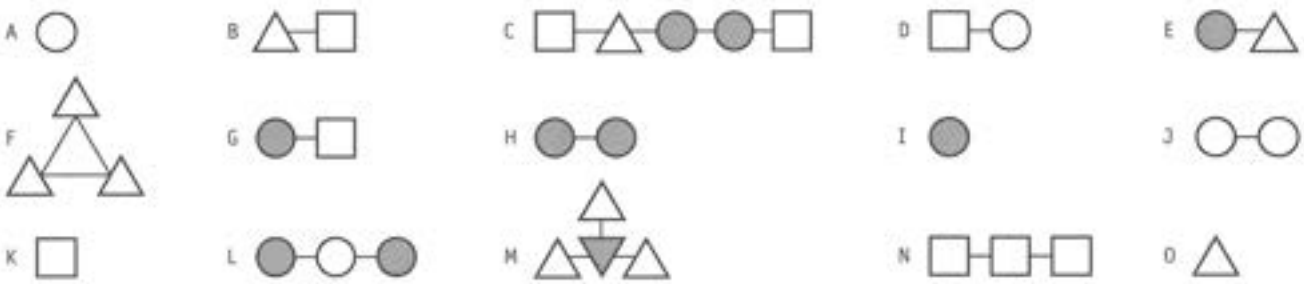
Atoms and molecules 3

Matter can be divided into groups using certain criteria. Matter can be divided into living and non-living things. Matter can be divided into solids, liquids and gases. Matter can be grouped into metals and non-metals. Matter can be split into single particles and multiple particles. Matter can be divided into substances which contain one kind of atom or more than one kind of atom.

Definitions

- **Atoms** are tiny particles. They are the simplest form of matter. Atoms are made up of single particles.
- **Molecules** are also tiny particles. They are made up of two or more atoms. These atoms are held together by chemical bonds.
- **Elements** are substances which contain only one kind of atom.
- **Compounds** are substances which contain particles that are made up of two or more different atoms.

Look at the following diagrams showing various particles and substances. Then answer the questions below by referring to the definitions.



1 Identify all atoms and all molecules and record them in the table.

Particles	Atoms	Molecules
Diagrams		

2 Identify all elements and all compounds and record them in the table.

Substances	Elements	Compounds
Diagrams		

3 Complete the following table about particles and substances by adding as many examples of each as possible from the diagrams.

Atom of element	Atom of compound	Molecule of element	Molecule of compound

4 Explain why one cell in the table from question 3 is left empty. _____

_____

5 Make up two more examples for each cell in question 3 and draw them below the table.

6 Explain what the lines in the diagrams represent. _____

_____

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